Consensus-Supervised U-Nets with PQ-Optimal Instance Selection for Solar Filament Segmentation

A Solution to the Solar Filament Segmentation Challenge 2026

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ABSTRACT

This report describes our solution to the Solar Filament Segmentation Challenge 2026, a Kaggle competition on automatic segmentation of solar filaments in GONG H-α observations. Five U-Nets with an ImageNet-pretrained EfficientNet-B3 encoder, one per date-grouped cross-validation fold, are trained on native-resolution crops of the 2048×2048 images with *soft consensus* targets (the fraction of annotators who marked each pixel). Their averaged probability map is converted into instances by a decision layer adapted from a public competition notebook: the map is thresholded at 16 levels, the nested connected components form a candidate forest, a gradient-boosted model predicts for every candidate the expected matched IoU against a random annotator, and an exact dynamic programme selects the pixel-disjoint subset that maximises the competition's pooled Panoptic Quality (PQ). Against the plain rule (threshold and minimum area), the decision layer raises out-of-fold pooled PQ from 0.405 to 0.431 on all 707 training images, with a gain on every fold, and the public leaderboard score from 0.35 to 0.37. Code, trained weights and a notebook reproducing the submission are public.[1]

1 INTRODUCTION

This report goes over the details of our submitted solution to the Solar Filament Segmentation Challenge 2026 [2]. The challenge, launched on July 10, 2026, is organized by the Earth-Space AI Research (ESAIR) Lab[2]. This challenge utilizes the MAGFiLO dataset for evaluation of the segmentation algorithms [1].

The task is *instance* segmentation evaluated with Panoptic Quality [8]: every predicted filament counts either as a true positive (IoU $>$ 0.5 with one ground-truth filament), or as a false positive, and each pixel may belong to at most one prediction. Three properties of the data shaped our design. First, the 707 training images carry 1154 annotation sets, because 296 images were annotated independently by two or three annotators who disagree noticeably on filament extent and on faint filaments. Second, filaments are small and thin (median area 1228 px, median bounding box ≈75×70 px in a 2048×2048 frame), so down-sampling destroys the barbs the organizers care about. Third, only 0.9% of the 8199 annotated filaments touch another filament within 3 px, so the instance problem is almost entirely a *semantic* segmentation problem followed by connected-component labelling, with a small residual need to split touching pairs. We therefore pair a semantic U-Net [9] with an explicit, metric-driven decision about *which* components to submit, and measure every design choice on out-of-fold data.

[1] https://github.com/cloudforge-oy/solar-filament-2026
[2] www.esairlab.com/

Kaggle Competition, 2026, kaggle.com/competitions/filament-segmentation-2026
2026.

2 METHODOLOGY

2.1 Data preparation and targets

Images are read as single-channel 8-bit grayscale and scaled to $[0, 1]$; no other pre-processing (no limb-darkening correction, no disk masking) is applied. For every annotation set, the single-polygon segmentations are rasterised with `pycocotools` into a 2048×2048 instance map. The k annotation sets of one image are then reduced to three *consensus* targets that are shared by all annotators of that image:

(1) the soft filament mask $y_{\text{seg}} \in [0, 1]$, the fraction of annotators who marked the pixel. Training on this soft label rather than on one arbitrary annotator (or on the union) makes the network predict the *expected* annotation, which is the right target when the hidden test ground truth also consists of several annotator sets;
(2) a normalised distance-to-boundary map y_{dist}, computed per instance as the Euclidean distance transform divided by the instance maximum, averaged over annotators;
(3) a contact-line map y_{touch}: pixels within 2 px of two different instances of the same annotation set, averaged over annotators.

The auxiliary targets were meant to separate touching filaments, but a distance-map watershed [3] over-segments elongated filaments even on ground truth (PQ 0.75 vs 0.79 for plain components) and the contact head, with only ≈70 touching pairs to learn from, never changed validation PQ. Both are kept as regularising auxiliary tasks; their predicted maps also serve as candidate descriptors (Section 2.4).

2.2 Model

We use the U-Net implementation of `segmentation_models_pytorch` [7] with ImageNet-pretrained encoders, the first convolution adapted to one input channel, and three output channels (filament logit, distance logit, contact logit). Two encoders were compared: ResNet34 [6] (24.4 M parameters) and EfficientNet-B3 [10] (13.2 M parameters); the final submission uses EfficientNet-B3 only. The loss is

$$\mathcal{L} = \text{BCE}(\hat{y}_{\text{seg}}, y_{\text{seg}}) + \text{Dice}(\hat{y}_{\text{seg}}, y_{\text{seg}}) + 5\,\text{MSE}(\hat{y}_{\text{dist}}, y_{\text{dist}}) + 0.5\,\text{BCE}_{w=20}(\hat{y}_{\text{touch}}, y_{\text{touch}}),$$

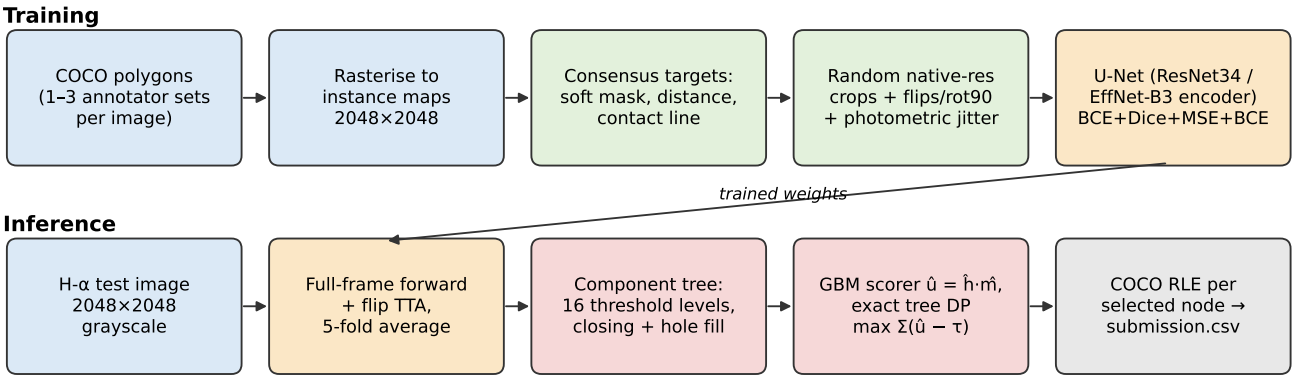


Figure 1: End-to-end pipeline. Training (top): COCO polygons of all annotator sets are rasterised into instance maps and reduced to three consensus targets per image; U-Nets are trained on random native-resolution crops. Inference (bottom): full-frame prediction with flip TTA averaged over the five fold models, a 16-level component tree, a learned score per candidate and an exact selection that maximises pooled PQ, COCO RLE encoding.

where both BCE terms accept soft targets and the contact term uses a positive-class weight of 20 because contact pixels are extremely rare.

2.3 Sampling and training

Networks are trained on random crops at native resolution (512×512 for ResNet34, 640×640 for EfficientNet-B3), four crops per image per epoch; with probability 0.7 the crop is centred near a filament pixel of the consensus mask, otherwise it is placed uniformly at random (so that empty sky, limb and off-disk corners are seen). Augmentations are horizontal/vertical flips, 90° rotations, random gamma ($e^{\pm 0.3}$), contrast (±15%), brightness (±0.05) and Gaussian noise ($\sigma \le 0.02$). We use AdamW (weight decay 10^{-4}), a one-cycle learning-rate schedule peaking at $3 \cdot 10^{-4}$, mixed precision, batch size 8 (ResNet34, 40 epochs) or 6 (EfficientNet-B3, 50 epochs). One ResNet34 run takes about 45 minutes on a laptop RTX 5060 (8 GB).

Validation split. The 707 images are grouped by observation date and the dates are assigned to five folds, so that the same day observed by two GONG stations, or the same image annotated by three annotators, never straddles the training/validation boundary. Every configuration is trained five times, once per held-out fold, so that each of the 1154 annotation sets receives an out-of-fold prediction; all numbers in Section 3 are out-of-fold. The final ensemble averages the five EfficientNet-B3 fold models (each trained on 80% of the images), which uses all training data without ever tuning on data a model has seen.

2.4 Inference and PQ-optimal instance selection

Each test image is processed as a whole 2048×2048 frame (0.4 s per forward pass in fp16); predictions are averaged over the identity and three flips (TTA) and over the five fold models. The conversion of this map p into instances follows the decision layer of a public competition notebook [11], which we re-implemented on our own maps (`src/decision.py`). The host pools TP, FP, FN and matched IoUs over all (image, annotator record) pairs, so

$$\mathrm{PQ}(P) = \frac{\sum_i R_i \sum_{c \in P_i} u_i(c)}{\frac{1}{2}\sum_i R_i |P_i| + \frac{1}{2} N_G}, \quad u_i(c) = \frac{1}{R_i} \sum_r \mathrm{IoU}(c, g_r)\, \mathbf{1}\{\mathrm{IoU} > 0.5\},$$

where R_i is the number of records of image i and u the *annotator-expected matched IoU* of candidate c. By Dinkelbach's theorem [4] the optimal submission maximises $\sum_c (u(c) - \tau)$ with a single global $\tau = \mathrm{PQ}^*/2$. **Candidates.** p is thresholded at 16 levels $(0.05, 0.10, \ldots, 0.80)$, each mask is closed (5×5 ellipse) and hole-filled; both operations are increasing, so the 8-connected components of all levels (≥100 px) form an exactly nested forest (≈170 nodes per image). Every antichain of this forest is a valid pixel-disjoint submission, and every single-threshold rule is one of them. **Scorer.** 35 descriptors per node (level, area, shape, solidity and elongation, statistics of p, of the distance and contact maps, relative darkness against a 40-px Gaussian background and against a surrounding ring, limb distance, tree relations, image activity, station) feed two gradient-boosted tree models [5] (scikit-learn): $\hat{h}$, the probability that a record matches the node, trained on (node, record) rows, and $\hat{m}$, the expected IoU given a match; $\hat{u} = \hat{h}\,\hat{m}$. For validation the scorer of fold k is trained only on candidates of the other four folds (cross-fitting); the submitted scorer uses all 119 802 out-of-fold candidates. **Selection.** $\max \sum_{c \in A} (\hat{u}(c) - \tau)$ over antichains A is solved exactly by $\mathrm{best}(v) = \max\{0, \hat{u}(v) - \tau, \sum_{w \in \mathrm{ch}(v)} \mathrm{best}(w)\}$ (verified against brute force). Selected nodes are disjoint by construction, as the scorer requires, and are encoded as COCO RLE.

3 EVALUATION

3.1 Strategy

We re-implemented PQ exactly as defined by the organizers: predicted and ground-truth instances are matched by IoU > 0.5 (unique by construction), $\mathrm{PQ} = \sum_{TP} \mathrm{IoU} / (|TP| + \frac{1}{2}|FP| + \frac{1}{2}|FN|)$, against every annotation set of an image. Our primary number is *pooled*

PQ, which sums the terms over all 1154 (image, record) pairs as the host does; Table 2 also gives the mean of per-set PQ with its standard error, and the consensus-mask pixel Dice.

3.2 Results

Table 1 shows the decision layer on the out-of-fold EfficientNet-B3 maps. Choosing the best single tree level changes nothing (0.405 to 0.406), so the gain does not come from the closing or the extra levels. The learned selection adds +0.027 pooled PQ, positive on every fold (+0.019 to +0.035). It keeps fewer, better candidates: false positives fall by a third (4051 to 2735) for 125 fewer true positives. Pooled PQ is flat for $\tau \in [0.26, 0.32]$ (Figure 2). On the test set, the averaged five-model maps are slightly less confident than single-model out-of-fold maps (5.9 vs 6.4 selected filaments per image at $\tau = 0.30$); $\tau = 0.26$ restores the out-of-fold selection rate and lies on the plateau, and was submitted. A candidate oracle that knows the true u reaches 0.551, so most of the remaining loss is in deciding which faint filaments annotators draw.

Table 2 compares the encoders with the plain connected-component rule, its threshold and minimum area re-optimised per fold. EfficientNet-B3 beats ResNet34 by 0.02–0.03 on every fold and averaging the two encoders does not help, so the submission uses EfficientNet-B3 only. The contact-line cut, the distance-map watershed and filtering components by peak or mean probability never improved PQ.

Table 1: Pooled out-of-fold PQ per fold (EfficientNet-B3 maps): plain rule ($p > 0.6$, area ≥ 400), best single level of the component tree, the decision layer ($\tau = 0.26$), and the selection with the true u.

Fold	CC rule	best tree level	decision layer	Δ	oracle u
0	0.417	0.419	**0.436**	+0.019	0.553
1	0.389	0.388	**0.423**	+0.035	0.543
2	0.421	0.421	**0.446**	+0.025	0.560
3	0.398	0.400	**0.429**	+0.031	0.545
4	0.402	0.404	**0.424**	+0.022	0.556
all	0.405	0.406	**0.431**	+0.027	0.551

On the public leaderboard (about half of the 180 test images, two decimals) the submissions scored 0.33 (single-channel ResNet34), 0.35 (EfficientNet-B3, one fold or five folds, plain rule) and **0.37** with the decision layer ($\tau = 0.26$ and 0.30 alike).

3.3 Discussion

Strengths. The pipeline is small (five 13 M-parameter U-Nets and two boosted-tree models), each U-Net trains in about 50 minutes on an RTX 4080, inference including the decision layer takes ≈6 s per image on a laptop, every submitted mask is a nested component of a calibrated probability map, so instances never overlap, and every component of the pipeline is backed by paired out-of-fold evidence. Consensus supervision makes p an estimate of annotator agreement, which is what the decision layer needs.

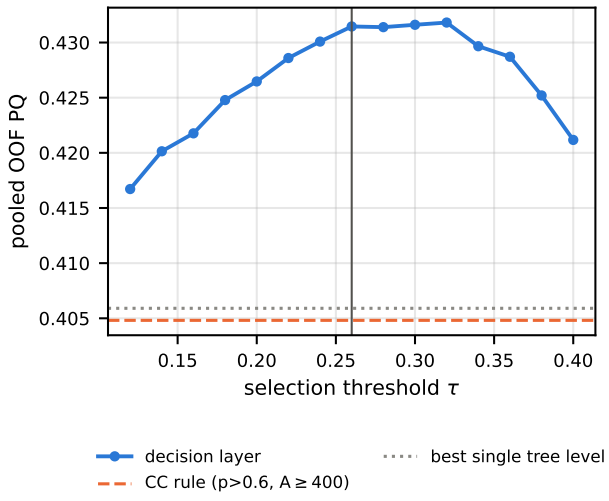


Figure 2: Pooled out-of-fold PQ of the decision layer against the selection threshold τ (all 707 images, cross-fitted scorer), with the plain rule and the best single tree level. The vertical line marks the submitted $\tau = 0.26$.

Weaknesses. The selection decides *whether* a candidate is submitted, not its outline: instances are level sets of one map, so a filament whose correct extent is not a level set (top row of Figure 3) cannot be recovered. False positives remain frequent because many of them are filaments that one annotator drew and another did not. Every honest pipeline we could inspect, including ours, loses ≈0.06 between out-of-fold and public leaderboard PQ (0.431 to 0.37 here). This happens for models of very different strength, which points to the composition and annotators of the ≈90 public test images rather than overfitting.

Possible improvements. An instance-conditioned refiner that re-segments each selected filament in a rotated crop (reported to add ≈0.01 in [11]), more diverse dense ensemble members and detector evidence as scorer descriptors, and modelling individual annotator styles.

4 ACKNOWLEDGMENT

The organization of the Filament Segmentation Challenge 2026 is partially supported by the U.S. National Science Foundation (NSF) under Grant No. 2209912 and 2433781, directorate for Computer and Information Science and Engineering (CSE), and Office of Advanced Cyberinfrastructure (OAC), and Grant No. 2511630, AST Division Of Astronomical Sciences and MPS Directorate for Mathematical and Physical Sciences.

The Filament Segmentation Challenge 2026 is sponsored by the U.S. National Science Foundation (NSF) National Solar Observatory (NSO).

This work utilizes GONG data obtained by the NSO Integrated Synoptic Program, managed by the National Solar Observatory, which is operated by the Association of Universities for Research in Astronomy (AURA), Inc. under a cooperative agreement with the

Table 2: Encoder comparison with the plain connected-component rule: out-of-fold PQ (mean over annotation sets) per fold and over all 1154 sets (± standard error), consensus-mask pixel Dice and matched counts. Flip TTA; threshold and minimum area re-optimised per fold.

Model	fold 0	fold 1	fold 2	fold 3	fold 4	pooled PQ	Dice	TP / FP / FN
ResNet34 U-Net	0.386	0.354	0.398	0.382	0.373	0.378 ± 0.004	0.660	5001 / 4782 / 3198
EfficientNet-B3 U-Net	0.409	0.382	0.426	0.405	0.399	0.404 ± 0.004	0.681	5115 / 3702 / 3084
Both encoders (equal weights)	0.407	0.373	0.419	0.405	0.395	0.399 ± 0.005	0.677	5122 / 3998 / 3077

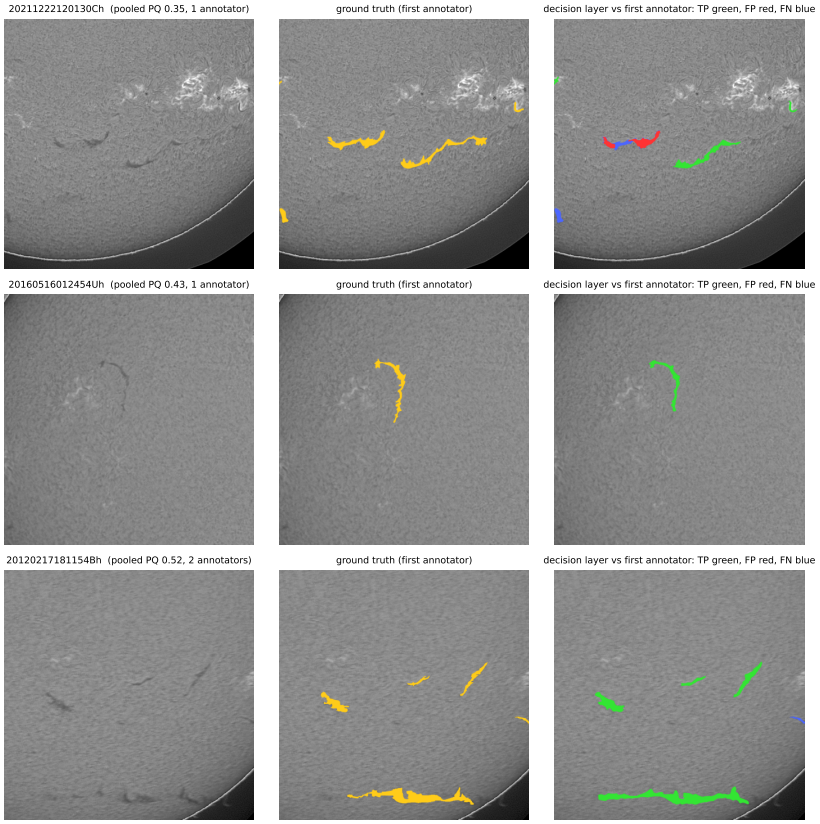


Figure 3: Fold-0 validation images at the 25th, 50th and 75th percentile of per-image pooled PQ (out-of-fold maps and scorer, $\tau = 0.26$). Left: H-α crop; middle: the first annotator's ground truth; right: selected instances coloured against that annotator (true positive green, false positive red, missed filament blue). In the top row one filament is submitted as two pieces, and neither piece reaches IoU 0.5.

National Science Foundation and with contribution from the National Oceanic and Atmospheric Administration. The GONG network of instruments is hosted by the Big Bear Solar Observatory, High Altitude Observatory, Learmonth Solar Observatory, Udaipur Solar Observatory, Instituto de Astrofísica de Canarias, and Cerro Tololo Interamerican Observatory.

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